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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Wave Optics

JEE Main Physics · Previous Year Questions with Answer Key

201	161	2002 – 2026	1	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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Numerical	52 questions
MCQ (Single Correct Answer)	148 questions
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Answer key	at the end

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Numerical 52 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Some distant star is to be observed by some telescope of diameter of objective lens a, at an angular resolution of 3.0×10^{-7} radian. If the wavelength of light from the star reaching the telescope is 500 nm , the minimum diameter of the objective lens of the telescope is ____ cm. (nearest interger)

Open solution

Q2. JEE Main 2026 (Online) 6th April Morning Shift

An unpolarized light of intensity I_0 passes through polarizer and then through a certain optically active solution and finally it goes to analyser. If the angle between analyser and polariser is 0° and intensity of light emerged from analyser is $\frac{3}{8} I_0$, the angle of rotation of the light by the solution with respect to analyser is ____ degrees.

Open solution

Q3. JEE Main 2026 (Online) 4th April Evening Shift

In a double slit experiment, when one of the slits is covered by a transparent mica sheet of refractive index 1.56 , the central fringe shifts to the position of 7th bright fringe, obtained with both slits uncovered. If the light source wavelength is 450 nm , the thickness of mica sheet is $\alpha \times 10^{-9}$ m. The value of α is

____ .

Open solution

Q4. JEE Main 2026 (Online) 2nd April Evening Shift

In a Young's double slit experiment, the intensity at some point on the screen is found to be $\frac{3}{4}$ times of the maximum of the interference pattern. The path difference between the interfering waves at this point is $\frac{\lambda}{x}$ where λ is wavelength of the incident light. The value of x is _____.

Open solution

Q5. JEE Main 2026 (Online) 2nd April Morning Shift

In single slit diffraction pattern, the wavelength of light used is 628 nm and slit width is 0.2 mm, the angular width of central maximum is $\alpha \times 10^{-2}$ degrees. The value of α is _____.

Open solution

Q6. JEE Main 2026 (Online) 28th January Evening Shift

A beam of light consisting of wavelengths 650 nm and 550 nm illuminates the Young's double slits with separation of 2 mm such that the interference fringes are formed on a screen, placed at a distance of 1.2 m from the slits. The least distance of a point from the central maximum, where the bright fringes due to both the wavelengths coincide, is _____ $\times 10^{-5}$ m.

Open solution

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Q7. JEE Main 2026 (Online) 23rd January Morning Shift

In two separate Young's double-slit experimental set-ups and two monochromatic light sources of different wavelengths are used to get fringes of equal width. The ratios of the slits separations and that of the wavelengths of light used are $2:1$ and $1:2$ respectively. The corresponding ratio of the distances between the slits and the respective screens (D_1/D_2) is ____。

Open solution

Q8. JEE Main 2026 (Online) 21st January Evening Shift

In a Young's double slit experiment set up, the two slits are kept 0.4 mm apart and screen is placed at 1 m from slits. If a thin transparent sheet of thickness $20\mu\,\mathrm{m}$ is introduced in front of one of the slits then center bright fringe shifts by 20 mm on the screen.

The refractive index of transparent sheet is given by $\frac{\alpha}{10}$, where α is ____.

Open solution

Q9. JEE Main 2025 (Online) 4th April Evening Shift

In a Young's double slit experiment, two slits are located 1.5 mm apart. The distance of screen from slits is 2 m and the wavelength of the source is 400 nm . If the 20 maxima of the double slit pattern are contained within the central maximum of the single slit diffraction pattern, then the width of each slit is $x \times 10^{-3}\,\mathrm{cm}$, where x-value is _____ .

Open solution

Q10. JEE Main 2025 (Online) 3rd April Morning Shift

Two coherent monochromatic light beams of intensities 4I and 9I are superimposed. The difference between the maximum and minimum intensities in the resulting interference pattern is $x\mathrm{I}$. The value of x is_____ .

Open solution

Q11. JEE Main 2025 (Online) 2nd April Morning Shift

If the measured angular separation between the second minimum to the left of the central maximum and the third minimum to the right of the central maximum is 30° in a single slit diffraction pattern recorded using 628 nm light, then the width of the slit is _____ μm.

Open solution

Q12. JEE Main 2025 (Online) 28th January Evening Shift

A thin transparent film with refractive index 1.4 , is held on circular ring of radius 1.8 cm . The fluid in the film evaporates such that transmission through the film at wavelength 560 nm goes to a minimum every 12 seconds. Assuming that the film is flat on its two sides, the rate of evaporation is _____ $\pi \times 10^{-13}\,\mathrm{m}^3/\mathrm{s}$.

Open solution

Q13. JEE Main 2025 (Online) 28th January Morning Shift

A double slit interference experiment performed with a light of wavelength 600 nm forms an interference fringe pattern on a screen with 10 th bright fringe having its centre at a distance of 10 mm from the central maximum. Distance of the centre of the same 10 th bright fringe from the central maximum when the source of light is replaced by another source of wavelength 660 nm would be _____ mm .

Open solution

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Q14. JEE Main 2024 (Online) 9th April Evening Shift

Monochromatic light of wavelength $500\ \text{nm}$ is used in Young's double slit experiment. An interference pattern is obtained on a screen. When one of the slits is covered with a very thin glass plate (refractive index $= 1.5$), the central maximum is shifted to a position previously occupied by the 4th bright fringe. The thickness of the glass-plate is _____ μm.

Open solution

Q15. JEE Main 2024 (Online) 9th April Morning Shift

In a Young's double slit experiment, the intensity at a point is $\left(\frac{1}{4}\right)^{\text{th}}$ of the maximum intensity, the minimum distance of the point from the central maximum is _____ μm. (Given : $\lambda = 600\ \text{nm},\ \text{d} = 1.0\ \text{mm}, \text{D} = 1.0\ \text{m}$)

Open solution

Q16. JEE Main 2024 (Online) 8th April Evening Shift

Two slits are $1\ \text{mm}$ apart and the screen is located $1\ \text{m}$ away from the slits. A light of wavelength $500\ \text{nm}$ is used. The width of each slit to obtain 10 maxima of the double slit pattern within the central maximum of the single slit pattern is _____ $\times 10^{-4}\ \text{m}$.

Open solution

Q17. JEE Main 2024 (Online) 8th April Morning Shift

A parallel beam of monochromatic light of wavelength $600\ \text{nm}$ passes through single slit of $0.4\ \text{mm}$ width. Angular divergence corresponding to second order minima would be _____ $\times 10^{-3}\ \text{rad}$.

Open solution

Q18. JEE Main 2024 (Online) 6th April Evening Shift

Two coherent monochromatic light beams of intensities I and $4\ \text{I}$ are superimposed. The difference between maximum and minimum possible intensities in the resulting beam is $x\ \text{I}$. The value of x is _____.

Open solution

Q19. JEE Main 2024 (Online) 5th April Evening Shift

In a single slit experiment, a parallel beam of green light of wavelength $550\ \text{nm}$ passes through a slit of width $0.20\ \text{mm}$. The transmitted light is collected on a screen $100\ \text{cm}$ away. The distance of first order minima from the central maximum will be $x \times 10^{-5}\ \text{m}$. The value of x is :

Open solution

Q20. JEE Main 2024 (Online) 5th April Morning Shift

In Young's double slit experiment, carried out with light of wavelength $5000\ \overset{o}{A}$, the distance between the slits is $0.3\ \text{mm}$ and the screen is at $200\ \text{cm}$ from the slits. The central maximum is at $x = 0\ \text{cm}$. The value of x for third maxima is _____ mm.

Open solution

Q21. JEE Main 2024 (Online) 4th April Morning Shift

Two wavelengths λ_1 and λ_2 are used in Young's double slit experiment. $\lambda_1 = 450\ \text{nm}$ and $\lambda_2 = 650\ \text{nm}$. The minimum order of fringe produced by λ_2 which overlaps with the fringe produced by λ_1 is n. The value of n is _____.

Open solution

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Q22. JEE Main 2024 (Online) 1st February Evening Shift

In Young's double slit experiment, monochromatic light of wavelength 5000 Å is used. The slits are $1.0\ \text{mm}$ apart and screen is placed at $1.0\ \text{m}$ away from slits. The distance from the centre of the screen where intensity becomes half of the maximum intensity for the first time is _____ $\times 10^{-6}$ m.

Open solution

Q23. JEE Main 2024 (Online) 31st January Morning Shift

Two waves of intensity ratio $1:9$ cross each other at a point. The resultant intensities at that point, when (a) Waves are incoherent is I_1 (b) Waves are coherent is I_2 and differ in phase by 60°. If $\frac{I_1}{I_2} = \frac{10}{x}$ then $x =$ _____.

Open solution

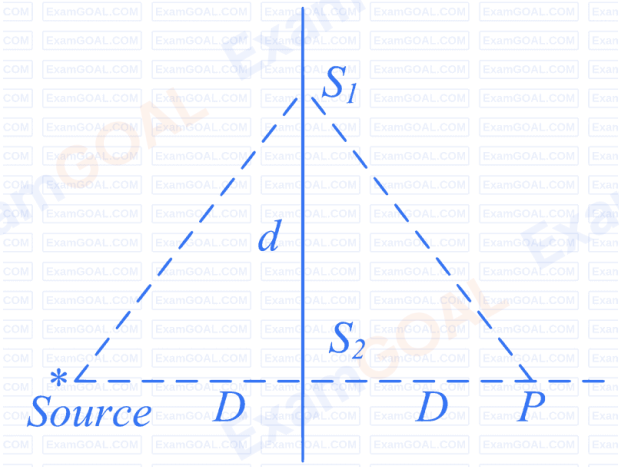
Q24. JEE Main 2024 (Online) 29th January Evening Shift

In a single slit diffraction pattern, a light of wavelength 6000$\overset{o}{A}$ is used. The distance between the first and third minima in the diffraction pattern is found to be $3\ \text{mm}$ when the screen in placed $50\ \text{cm}$ away from slits. The width of the slit is _____ $\times 10^{-4}\ \text{m}$.

Open solution

Q25. JEE Main 2024 (Online) 29th January Morning Shift

In a double slit experiment shown in figure, when light of wavelength $400\ \text{nm}$ is used, dark fringe is observed at P. If $\text{D} = 0.2\ \text{m}$, the minimum distance between the slits S_1 and S_2 is _____ mm.



Open solution

Q26. JEE Main 2024 (Online) 27th January Evening Shift

A parallel beam of monochromatic light of wavelength 5000 $\overset{o}{A}$ is incident normally on a single narrow slit of width $0.001\ \text{mm}$. The light is focused by convex lens on screen, placed on its focal plane. The first minima will be formed for the angle of diffraction of _____ (degree).

Open solution

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Q27. JEE Main 2023 (Online) 10th April Morning Shift

Unpolarised light of intensity 32 Wm^{-2} passes through the combination of three polaroids such that the pass axis of the last polaroid is perpendicular to that of the pass axis of first polaroid. If intensity of emerging light is 3 Wm^{-2}, then the angle between pass axis of first two polaroids is _____ $^{\circ}$.

Open solution

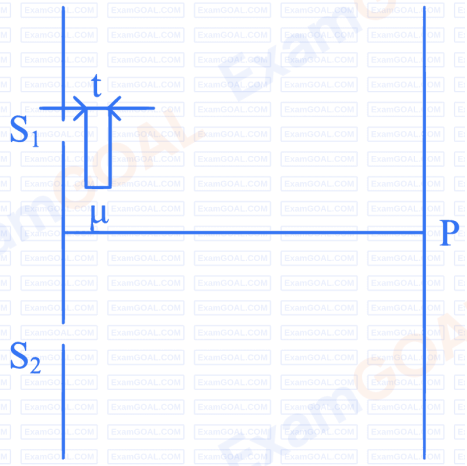
Q28. JEE Main 2023 (Online) 6th April Evening Shift

A beam of light consisting of two wavelengths $7000\ \overset{o}{A}$ and $5500\ \overset{o}{A}$ is used to obtain interference pattern in Young's double slit experiment. The distance between the slits is $2.5\ \text{mm}$ and the distance between the plane of slits and the screen is $150\ \text{cm}$. The least distance from the central fringe, where the bright fringes due to both the wavelengths coincide, is $n \times 10^{-5}\ \text{m}$. The value of n is _____.

Open solution

Q29. JEE Main 2023 (Online) 1st February Evening Shift

As shown in the figure, in Young's double slit experiment, a thin plate of thickness $t = 10\mu\text{m}$ and refractive index $\mu = 1.2$ is inserted in front of slit S_1. The experiment is conducted in air $(\mu = 1)$ and uses a monochromatic light of wavelength $\lambda = 500\ \text{nm}$. Due to the insertion of the plate, central maxima is shifted by a distance of $x\beta_0$. β_0 is the fringe-width before the insertion of the plate. The value of the x is _____.



Open solution

Q30. JEE Main 2023 (Online) 31st January Evening Shift

Two light waves of wavelengths 800 and $600\ \text{nm}$ are used in Young's double slit experiment to obtain interference fringes on a screen placed $7\ \text{m}$ away from plane of slits. If the two slits are separated by $0.35\ \text{mm}$, then shortest distance from the central bright maximum to the point where the bright fringes of the two wavelength coincide will be _____ mm.

Open solution

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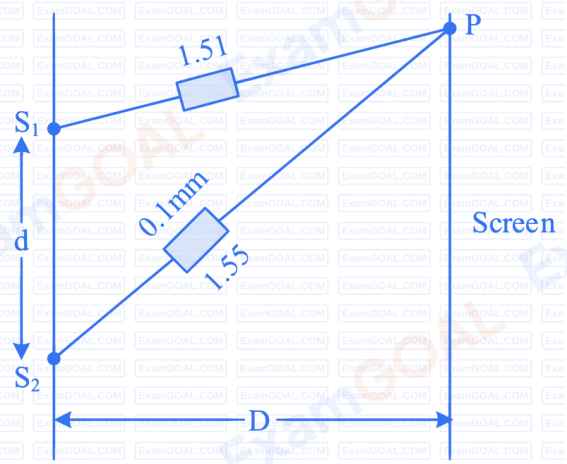
Q31. JEE Main 2023 (Online) 30th January Evening Shift

In a Young's double slit experiment, the intensities at two points, for the path differences $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$ (λ being the wavelength of light used) are I_1 and I_2 respectively. If I_0 denotes the intensity produced by each one of the individual slits, then $\frac{I_1+I_2}{I_0}$ = _____.

Open solution

Q32. JEE Main 2023 (Online) 30th January Morning Shift

In Young's double slit experiment, two slits S_1 and S_2 are 'd' distance apart and the separation from slits to screen is D (as shown in figure). Now if two transparent slabs of equal thickness 0.1 mm but refractive index 1.51 and 1.55 are introduced in the path of beam ($\lambda = 4000\ \overset{o}{A}$) from S_1 and S_2 respectively. The central bright fringe spot will shift by _____ number of fringes.



Open solution

Q33. JEE Main 2023 (Online) 29th January Evening Shift

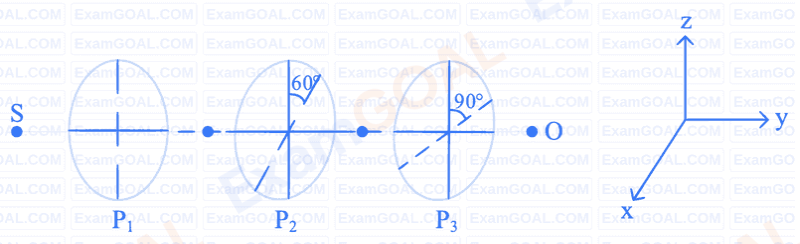
Unpolarised light is incident on the boundary between two dielectric media, whose dielectric constants are 2.8 (medium -1) and 6.8 (medium -2), respectively. To satisfy the condition, so that the reflected and refracted rays are perpendicular to each other, the angle of incidence should be $\tan^{-1}\left(1+\frac{10}{\theta}\right)^{\frac{1}{2}}$ the value of θ is _____.

(Given for dielectric media, $\mu_r = 1$)

Open solution

Q34. JEE Main 2023 (Online) 29th January Morning Shift

As shown in the figure, three identical polaroids P_1, P_2 and P_3 are placed one after another. The pass axis of P_2 and P_3 are inclined at angle of 60° and 90° with respect to axis of P_1. The source S has an intensity of 256 $\frac{W}{m^2}$. The intensity of light at point O is _____ $\frac{W}{m^2}$.



Open solution

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Q35. JEE Main 2022 (Online) 29th July Morning Shift

Two light beams of intensities 4I and 9I interfere on a screen. The phase difference between these beams on the screen at point A is zero and at point B is π. The difference of resultant intensities, at the point A and B, will be _____ I.

Open solution

Q36. JEE Main 2022 (Online) 28th July Morning Shift

In a Young's double slit experiment, a laser light of 560 nm produces an interference pattern with consecutive bright fringes' separation of 7.2 mm. Now another light is used to produce an interference pattern with consecutive bright fringes' separation of 8.1 mm. The wavelength of second light is _____ nm.

Open solution

Q37. JEE Main 2022 (Online) 27th July Morning Shift

Two beams of light having intensities I and 4I interfere to produce a fringe pattern on a screen. The phase difference between the two beams are $\pi/2$ and $\pi/3$ at points A and B respectively. The difference between the resultant intensities at the two points is xI. The value of x will be _____.

Open solution

Q38. JEE Main 2022 (Online) 29th June Evening Shift

In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-3} cm. If the distance between the slits is 1 mm, then the wavelength of the light will be _____ nm.

Open solution

Q39. JEE Main 2022 (Online) 28th June Evening Shift

In a Young's double slit experiment, an angular width of the fringe is 0.35° on a screen placed at 2 m away for particular wavelength of 450 nm. The angular width of the fringe, when whole system is immersed in a medium of refractive index 7/5, is $\frac{1}{\alpha}$. The value of α is _____.

Open solution

Q40. JEE Main 2022 (Online) 27th June Morning Shift

In Young's double slit experiment the two slits are 0.6 mm distance apart. Interference pattern is observed on a screen at a distance 80 cm from the slits. The first dark fringe is observed on the screen directly opposite to one of the slits. The wavelength of light will be _____ nm.

Open solution

Q41. JEE Main 2022 (Online) 24th June Morning Shift

Sodium light of wavelengths 650 nm and 655 nm is used to study diffraction at a single slit of aperture 0.5 mm. The distance between the slit and the screen is 2.0 m. The separation between the positions of the first maxima of diffraction pattern obtained in the two cases is _____ $\times 10^{-5}$ m.

Open solution

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Q42. JEE Main 2021 (Online) 1st September Evening Shift

The width of one of the two slits in a Young's double slit experiment is three times the other slit. If the amplitude of the light coming from a slit is proportional to the slit-width, the ratio of minimum to maximum intensity in the interference pattern is x : 4 where x is _____.

Open solution

Q43. JEE Main 2021 (Online) 31st August Evening Shift

In a Young's double slit experiment, the slits are separated by 0.3 mm and the screen is 1.5 m away from the plane of slits. Distance between fourth bright fringes on both sides of central bright is 2.4 cm. The frequency of light used is _____ $\times 10^{14}$ Hz.

Open solution

Q44. JEE Main 2021 (Online) 26th August Evening Shift

A source of light is placed in front of a screen. Intensity of light on the screen is I. Two Polaroids P_1 and P_2 are so placed in between the source of light and screen that the intensity of light on screen is I/2. P_2 should be rotated by an angle of (degrees) so that the intensity of light on the screen becomes $\frac{3I}{8}$.

Open solution

Q45. JEE Main 2021 (Online) 26th August Morning Shift

White light is passed through a double slit and interference is observed on a screen 1.5 m away. The separation between the slits is 0.3 mm. The first violet and red fringes are formed 2.0 mm and 3.5 mm away from the central white fringes. the difference in wavelengths of red and violet light is ..... nm.

Open solution

Q46. JEE Main 2021 (Online) 27th July Evening Shift

The difference in the number of waves when yellow light propagates through air and vacuum columns of the same thickness is one. The thickness of the air column is _____ mm. [Refractive index of air = 1.0003, wavelength of yellow light in vacuum = 6000 $\overset{o}{A}$]

Open solution

Q47. JEE Main 2021 (Online) 18th March Evening Shift

A galaxy is moving away from the earth at a speed of 286 kms^{-1}. The shift in the wavelength of a redline at 630 nm is x $\times 10^{-10}$ m. The value of x, to the nearest integer, is _____. [Take the value of speed of light c, as 3 $\times 10^8$ ms^{-1}]

Open solution

Q48. JEE Main 2021 (Online) 16th March Morning Shift

A fringe width of 6 mm was produced for two slits separated by 1 mm apart. The screen is placed 10 m away. The wavelength of light used is 'x' nm. The value of 'x' to the nearest integer is _____.

Open solution

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Q49. JEE Main 2021 (Online) 24th February Morning Shift

An unpolarised light beam is incident on the polarizer of a polarization experiment and the intensity of light beam emerging from the analyzer is measured as 100 Lumens. Now, if the analyzer is rotated around the horizontal axis (direction of light) by 30° in clockwise direction, the intensity of emerging light will be _____ Lumens.

Open solution

Q50. JEE Main 2020 (Online) 6th September Evening Slot

A Young's double-slit experiment is performed using monochromatic light of wavelength λ. The intensity of light at a point on the screen, where the path difference is λ, is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{6}$ is given by $\frac{nK}{12}$, where n is an integer. The value of n is _____.

Open solution

Q51. JEE Main 2020 (Online) 4th September Evening Slot

Orange light of wavelength 6000×10^{-10} m illuminates a single slit of width 0.6×10^{-4} m. The maximum possible number of diffraction minima produced on both sides of the central maximum is _____.

Open solution

Q52. JEE Main 2020 (Online) 9th January Evening Slot

In a Young's double slit experiment 15 fringes are observed on a small portion of the screen when light of wavelength 500 nm is used. Ten fringes are observed on the same section of the screen when another light source of wavelength λ is used. Then the value of λ is (in nm) _____.

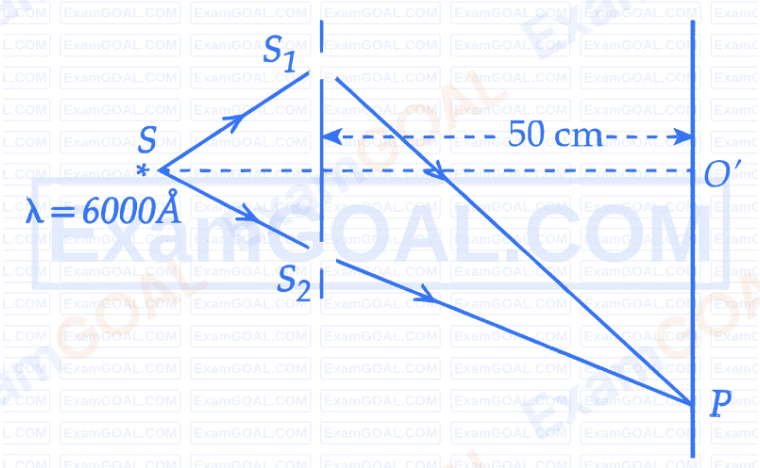
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MCQ (Single Correct Answer) 148 questions

Q53. JEE Main 2026 (Online) 6th April Evening Shift

In a Young double slit experiment, the wavelength of incident light is $6000\mathring{A}$, the separation between slits S_1 and S_2 is 5 cm and the distance between slits plane and screen is 50 cm , as shown in the figure below. If the resultant intensity at P is equal to the intensity due to individual slits, the path difference between interfering waves is ____ Å.



Open solution

(A) 4000 (B) 3000 (C) 2000 (D) 1000

Q54. JEE Main 2026 (Online) 6th April Morning Shift

In interference experiment the path difference between two interfering waves at a point A on the screen is $\lambda/3$, where λ is the wavelength of these waves, and at another point B the path difference is $\lambda/6$. The ratio of intensities at points A and B is ____ .

Open solution

(A) 3 (B) 4 (C) 1/3 (D) 1/4

Q55. JEE Main 2026 (Online) 5th April Evening Shift

The maximum intensity in a Young's double slit experiment is I_0. Distance between the slits (d) is 5λ, where λ is the wavelength of light used. The intensity of the fringe, exactly opposite to one of the slits on the screen, placed at $D = 10d$ is ____ .

Open solution

(A) $\frac{I_0}{4}$ (B) $\frac{I_o}{2}$ (C) I_o (D) $\frac{3I_0}{4}$

Q56. JEE Main 2026 (Online) 5th April Morning Shift

In Young's double slit experiment, the fringe width of the interference pattern produced on the screen is $2.4\mu\,\mathrm{m}$. If the experiment is carried out in another medium having refractive index 1.2 , the fringe width will be ____ $\mu\mathrm{m}$.

Open solution

(A) 1.2 (B) 2 (C) 2.4 (D) 2.88

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Q57. JEE Main 2026 (Online) 4th April Evening Shift

An unpolarized light of certain intensity passes through a combination of two polarizers whose transmission axes are at 30° and 90°, respectively, with respect to the horizontal axis. A third polarizer with its transmission axis at 60° with the horizontal axis is placed between the two existing polarizers. The ratio of the output intensities with and without the third polarizer is ____ .

Open solution

(A) 3/4 (B) 4/3 (C) 9/4 (D) 4/9

Q58. JEE Main 2026 (Online) 4th April Morning Shift

A slit of width a is illuminated by light of wavelength λ. The linear separation between 1st and 3rd minima in the diffraction pattern produced on a screen placed at a distance D from the slit system is ____ .

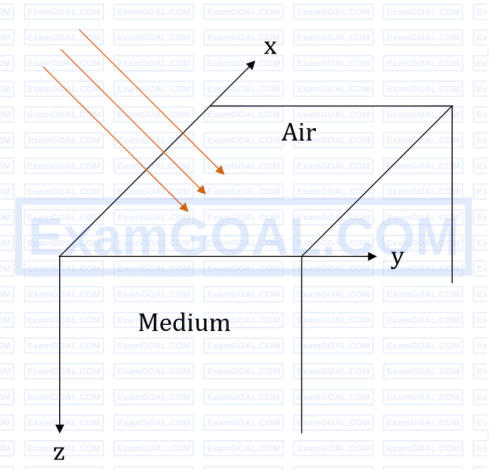
Open solution

(A) $\frac{D\lambda}{a}$ (B) $1.5\frac{D\lambda}{a}$ (C) $2\frac{D\lambda}{a}$ (D) $3\frac{D\lambda}{a}$

Q59. JEE Main 2026 (Online) 4th April Morning Shift

An unpolarized light is incident on the plane interface of air-dielectric medium shown in figure. If the incident angle is equal to Brewster angle, identify the expression representing reflected wave.

Open solution



(A) $\left(E_x\hat{i} + E_y\hat{j}\right)\sin(kx - kz - \omega t)$ (B) $\left(E_x\hat{i} + E_z\hat{k}\right)\sin(kx + ky - \omega t)$

(C) $\left(E_x\hat{j} + E_y\hat{k}\right)\sin(ky + kz - \omega t)$ (D) $\left(E_x\hat{i} + E_y\hat{j} + E_z\hat{k}\right)\sin(kx + ky - kz - \omega t)$

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Q60. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

Given below are two statements :

Statement I : A plane wave after passing through prism remains as plane wave but passing through small pin hole may become spherical wave.

Statement II : The curvature of a spherical wave emerging from a slit will increase for increasing slit width.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q61. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

In the Young's double slit experiment the intensity produced by each one of the individual slits is I_o. The distance between two slits is 2 mm . The distance of screen from slits is 10 m . The wavelength of light is $6000\ \mathrm{A}^\circ$. The intensity of light on the screen in front of one of the slits is ____

(A) $\frac{I_o}{2}$ (B) I_o (C) $2I_o$ (D) $4I_o$

Q62. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

When an unpolarized light falls at a particular angle on a glass plate (placed in air), it is observed that the reflected beam is linearly polarized. The angle of refracted beam with respect to the normal is ____ .

$(\tan^{-1}(1.52) = 57.7^\circ$, refractive indices of air and glass are 1.00 and 1.52, respectively.)

(A) 36.3° (B) 39.6° (C) 42.6° (D) 32.3°

Q63. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

The wavelength of light, while it is passing through water is 540 nm . The refractive index of water is $4/3$. The wavelength of the same light when it is passing through a transparent medium having refractive index of $3/2$ is ____ nm.

(A) 540 (B) 840 (C) 480 (D) 380

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Q64. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Which of the following are true for a single slit diffraction?

A. Width of central maxima increases with increase in wavelength keeping slit width constant.

B. Width of central maxima increases with decrease in wavelength keeping slit width constant.

C. Width of central maxima increases with decrease in slit width at constant wavelength.

D. Width of central maxima increases with increase in slit width at constant wavelength.

E. Brightness of central maxima increases for decrease in wavelength at constant slit width.

(A) B, D only (B) A, D only (C) A, C, E only (D) B, C only

Q65. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

Given below are two statements :

Statement I : In a Young's double slit experiment, the angular separation of fringes will increase as the screen is moved away from the plane of the slits

Statement II : In a Young's double slit experiment, the angular separation of fringes will increase when monochromatic source is replaced by another monochromatic source of higher wavelength

In the light of the above statements, choose the **correct answer** from the options given below :

(A) Both **Statement I** and **Statement II** are true

(B) **Statement I** is true but **Statement II** is false

(C) **Statement I** is false but **Statement II** is true

(D) Both **Statement I** and **Statement II** are false

Q66. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

In a double slit experiment the distance between the slits is 0.1 cm and the screen is placed at 50 cm from the slits plane. When one slit is covered with a transparent sheet having thickness t and refractive index $n(=1.5)$, the central fringe shifts by 0.2 cm . The value of t is

____ cm.

(A) 6.0×10^{-3} (B) 8×10^{-4} (C) 5.0×10^{-3} (D) 5.6×10^{-4}

Q67. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

In a Young's double slit experiment, the source is white light. One of the slits is covered by red filter and another by a green filter. In this case:

(A) there shall be alternate interference fringes of red and green.

(B) there shall be an interference pattern for red distinct from that for green.

(C) there shall be an interference pattern, where each fringe's pattern center is green and outer edges is red.

(D) there shall be no interference fringes.

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Q68. JEE Main 2025 (Online) 7th April Morning Shift

Two plane polarized light waves combine at a certain point whose electric field components are

$E_1 = E_0 \operatorname{Sin} \omega t$

$E_2 = E_0 \operatorname{Sin} \left(\omega t + \frac{\pi}{3}\right)$

Find the amplitude of the resultant wave.

Open solution

(A) $1.7\,\mathrm{E}_0$ (B) E_0 (C) $0.9\,\mathrm{E}_0$ (D) $3.4\,\mathrm{E}_0$

Q69. JEE Main 2025 (Online) 4th April Evening Shift

Two polarisers P_1 and P_2 are placed in such a way that the intensity of the transmitted light will be zero. A third polariser P_3 is inserted in between P_1 and P_2, at particular angle between P_2 and P_3. The transmitted intensity of the light passing through all three polarisers is maximum. The angle between the polarisers P_2 and P_3 is :

Open solution

(A) $\pi/6$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\pi/8$

Q70. JEE Main 2025 (Online) 4th April Morning Shift

In a Young's double slit experiment, the slits are separated by 0.2 mm . If the slits separation is increased to 0.4 mm , the percentage change of the fringe width is :

Open solution

(A) 25% (B) 50% (C) 0% (D) 100%

Q71. JEE Main 2025 (Online) 3rd April Evening Shift

Two monochromatic light beams have intensities in the ratio 1:9. An interference pattern is obtained by these beams. The ratio of the intensities of maximum to minimum is

Open solution

(A) $8:1$ (B) $4:1$ (C) $3:1$ (D) $9:1$

Q72. JEE Main 2025 (Online) 3rd April Evening Shift

Width of one of the two slits in a Young's double slit interference experiment is half of the other slit. The ratio of the maximum to the minimum intensity in the interference pattern is :

Open solution

(A) $3:1$ (B) $(2\sqrt{2}+1):(2\sqrt{2}-1)$

(C) $9:1$ (D) $(3+2\sqrt{2}):(3-2\sqrt{2})$

Q73. JEE Main 2025 (Online) 3rd April Evening Shift

A monochromatic light of frequency $5 \times 10^{14}\,\mathrm{Hz}$ travelling through air, is incident on a medium of refractive index ' 2 '. Wavelength of the refracted light will be :

Open solution

(A) 400 nm (B) 300 nm (C) 600 nm (D) 500 nm

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Q74. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

A light wave is propagating with plane wave fronts of the type $x + y + z =$ constant. The angle made by the direction of wave propagation with the x-axis is :

(A) $\cos^{-1}(2/3)$

(B) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

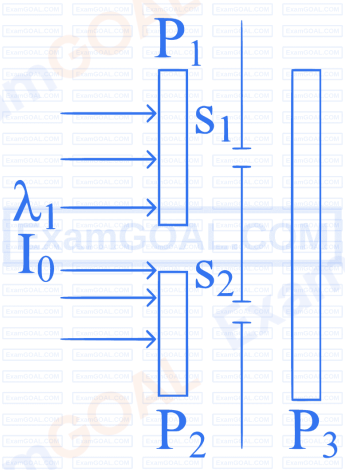
(C) $\cos^{-1}\left(\frac{1}{3}\right)$

(D) $\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Q75. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

In a Young's double slit experiment, three polarizers are kept as shown in the figure. The transmission axes of P_1 and P_2 are orthogonal to each other. The polarizer P_3 covers both the slits with its transmission axis at 45° to those of P_1 and P_2. An unpolarized light of wavelength λ and intensity I_0 is incident on P_1 and P_2. The intensity at a point after P_3 where the path difference between the light waves from s_1 and s_2 is $\frac{\lambda}{3}$, is



(A) I_0

(B) $\frac{I_0}{3}$

(C) $\frac{I_0}{4}$

(D) $\frac{I_0}{2}$

Q76. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Young's double slit interference apparatus is immersed in a liquid of refractive index 1.44. It has slit separation of 1.5 mm . The slits are illuminated by a parallel beam of light whose wavelength in air is 690 nm . The fringe-width on a screen placed behind the plane of slits at a distance of 0.72 m , will be:

(A) 0.63 mm

(B) 0.33 mm

(C) 0.46 mm

(D) 0.23 mm

Q77. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

The Young's double slit interference experiment is performed using light consisting of 480 nm and 600 nm wavelengths to form interference patterns. The least number of the bright fringes of 480 nm light that are required for the first coincidence with the bright fringes formed by 600 nm light is

(A) 8

(B) 4

(C) 5

(D) 6

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Q78. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

The width of one of the two slits in Young's double slit experiment is d while that of the other slit is x d. If the ratio of the maximum to the minimum intensity in the interference pattern on the screen is $9:4$ then what is the value of x ? (Assume that the field strength varies according to the slit width.)

(A) 2 (B) 3 (C) 4 (D) 5

Q79. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

A transparent film of refractive index, 2.0 is coated on a glass slab of refractive index, 1.45. What is the minimum thickness of transparent film to be coated for the maximum transmission of Green light of wavelength 550 nm . [Assume that the light is incident nearly perpendicular to the glass surface.]

(A) 68.7 nm (B) 137.5 nm (C) 94.8 nm (D) 275 nm

Q80. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In Young's double slit experiment, the fringes produced by red light are closer as compared to those produced by blue light.

Reason (R) : The fringe width is directly proportional to the wavelength of light.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both $(\mathbf{A})$ and $(\mathbf{R})$ are true but $(\mathbf{R})$ is NOT the correct explanation of (A)

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Q81. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion-(A) : If Young's double slit experiment is performed in an optically denser medium than air, then the consecutive fringes come closer.

Reason-(R) : The speed of light reduces in an optically denser medium than air while its frequency does not change.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) (A) is false but (R) is true

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) Both (A) and (R) are true and (R) is the correct explanation of (A)

(D) (A) is true but (R) is false

Q82. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

Light emerges out of a convex lens when a source of light is kept at its focus. The shape of wavefront of the light is :

(A) cylindrical

(B) spherical

(C) plane

(D) both spherical and cylindrical

Q83. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

The width of one of the two slits in a Young's double slit experiment is 4 times that of the other slit. The ratio of the maximum to the minimum intensity in the interference pattern is:

(A) $1:1$ **(B)** $4:1$ **(C)** $16:1$ **(D)** $9:1$

Q84. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

A microwave of wavelength $2.0\ \text{cm}$ falls normally on a slit of width $4.0\ \text{cm}$. The angular spread of the central maxima of the diffraction pattern obtained on a screen $1.5\ \text{m}$ away from the slit, will be :

(A) 60° **(B)** 45° **(C)** 15° **(D)** 30°

Q85. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

A monochromatic light of wavelength $6000\ \text{Å}$ is incident on the single slit of width $0.01\ \text{mm}$. If the diffraction pattern is formed at the focus of the convex lens of focal length $20\ \text{cm}$, the linear width of the central maximum is :

(A) $12\ \text{mm}$ **(B)** $24\ \text{mm}$ **(C)** $60\ \text{mm}$ **(D)** $120\ \text{mm}$

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Q86. JEE Main 2024 (Online) 31st January Evening Shift

When unpolarized light is incident at an angle of 60° on a transparent medium from air, the reflected ray is completely polarized. The angle of refraction in the medium is:

(A) 60° (B) 90° (C) 30° (D) 45°

Open solution

Q87. JEE Main 2024 (Online) 30th January Evening Shift

A beam of unpolarised light of intensity I_0 is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of 45° relative to that of A. The intensity of emergent light is:

(A) $I_0/2$ (B) $I_0/8$ (C) $I_0/4$ (D) I_0

Open solution

Q88. JEE Main 2024 (Online) 30th January Morning Shift

The diffraction pattern of a light of wavelength $400\ \text{nm}$ diffracting from a slit of width $0.2\ \text{mm}$ is focused on the focal plane of a convex lens of focal length $100\ \text{cm}$. The width of the 1st secondary maxima will be :

(A) 2 mm (B) 0.2 mm (C) 0.02 mm (D) 2 cm

Open solution

Q89. JEE Main 2024 (Online) 29th January Evening Shift

In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $7\lambda/4$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is :

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Open solution

Q90. JEE Main 2024 (Online) 27th January Evening Shift

When a polaroid sheet is rotated between two crossed polaroids then the transmitted light intensity will be maximum for a rotation of :

(A) 90° (B) 30° (C) 45° (D) 60°

Open solution

Q91. JEE Main 2023 (Online) 15th April Morning Shift

A single slit of width a is illuminated by a monochromatic light of wavelength $600\ \text{nm}$. The value of ' a ' for which first minimum appears at $\theta = 30^\circ$ on the screen will be :

(A) $3\mu\text{m}$ (B) $0.6\mu\text{m}$ (C) $1.8\mu\text{m}$ (D) $1.2\mu\text{m}$

Open solution

Q92. JEE Main 2023 (Online) 13th April Evening Shift

In a Young's double slits experiment, the ratio of amplitude of light coming from slits is $2:1$. The ratio of the maximum to minimum intensity in the interference pattern is:

(A) 25 : 9 (B) 9 : 1 (C) 9 : 4 (D) 2 : 1

Open solution

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Q93. JEE Main 2023 (Online) 10th April Evening Shift

The ratio of intensities at two points P and Q on the screen in a Young's double slit experiment where phase difference between two waves of same amplitude are $\pi/3$ and $\pi/2$, respectively are

(A) 2 : 3 (B) 1 : 3 (C) 3 : 1 (D) 3 : 2

Open solution

Q94. JEE Main 2023 (Online) 8th April Evening Shift

The width of fringe is $2\ \mathrm{mm}$ on the screen in a double slits experiment for the light of wavelength of $400\ \mathrm{nm}$. The width of the fringe for the light of wavelength 600 nm will be:

(A) 4 mm (B) 1.33 mm (C) 2 mm (D) 3 mm

Open solution

Q95. JEE Main 2023 (Online) 1st February Morning Shift

'n' polarizing sheets are arranged such that each makes an angle 45° with the preceding sheet. An unpolarized light of intensity I is incident into this arrangement. The output intensity is found to be $I/64$. The value of n will be:

(A) 4 (B) 5 (C) 3 (D) 6

Open solution

Q96. JEE Main 2023 (Online) 31st January Morning Shift

Two polaroids A and B are placed in such a way that the pass-axis of polaroids are perpendicular to each other. Now, another polaroid C is placed between A and B bisecting angle between them. If intensity of unpolarized light is I_0 then intensity of transmitted light after passing through polaroid B will be:

(A) $\frac{I_0}{4}$ (B) $\frac{I_0}{8}$ (C) Zero (D) $\frac{I_0}{2}$

Open solution

Q97. JEE Main 2023 (Online) 29th January Evening Shift

A scientist is observing a bacteria through a compound microscope. For better analysis and to improve its resolving power he should. (Select the best option)

(A) Decrease the focal length of the eye piece.

(B) Increase the wavelength of the light

(C) Increase the refractive index of the medium between the object and objective lens

(D) Decrease the diameter of the objective lens

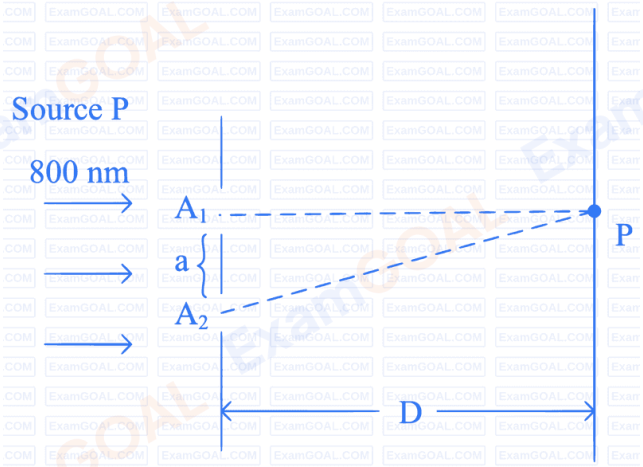
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Q98. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

In a Young's double slit experiment, two slits are illuminated with a light of wavelength $800\ \text{nm}$. The line joining A_1P is perpendicular to A_1A_2 as shown in the figure. If the first minimum is detected at P, the value of slits separation 'a' will be:



The distance of screen from slits D = 5 cm

(A) 0.2 mm (B) 0.5 mm (C) 0.4 mm (D) 0.1 mm

Q99. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

In Young's double slits experiment, the position of 5th bright fringe from the central maximum is 5 cm. The distance between slits and screen is 1 m and wavelength of used monochromatic light is 600 nm. The separation between the slits is :

(A) 60 μm (B) 48 μm (C) 36 μm (D) 12 μm

Q100. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

Given below are two statements :

Statement I : If the Brewster's angle for the light propagating from air to glass is θ_{B}, then the Brewster's angle for the light propagating from glass to air is $\frac{\pi}{2} - \theta_B$

Statement II : The Brewster's angle for the light propagating from glass to air is $\tan^{-1}(\mu_{\text{g}})$ where μ_{g} is the refractive index of glass.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

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Q101. JEE Main 2022 (Online) 30th July Evening Shift

Two poloroids P_1 and P_2 are placed parallel to each other with common axis.The light of intensity I_0 passes through a polaroid sheet P_1 and then passes through polaroid P_2. If P_2 is now rotated by 60°, then the intensity of the output light from P_2 will be:

(A) $\frac{I_0}{2}$ (B) $\frac{I_0}{4}$ (C) Zero (D) $\frac{I_0}{8}$

Open solution

Q102. JEE Main 2022 (Online) 29th July Evening Shift

An unpolarised light beam of intensity $2I_0$ is passed through a polaroid P and then through another polaroid Q which is oriented in such a way that its passing axis makes an angle of 30° relative to that of P. The intensity of the emergent light is

(A) $\frac{\mathrm{I}_0}{4}$ (B) $\frac{\mathrm{I}_0}{2}$ (C) $\frac{3I_0}{4}$ (D) $\frac{3\mathrm{I}_0}{2}$

Open solution

Q103. JEE Main 2022 (Online) 27th July Evening Shift

Two coherent sources of light interfere. The intensity ratio of two sources is $1:4$. For this interference pattern if the value of $\frac{I_{\max}+I_{\min}}{I_{\max}-I_{\min}}$ is equal to $\frac{2\alpha+1}{\beta+3}$, then $\frac{\alpha}{\beta}$ will be :

(A) 1.5 (B) 2 (C) 0.5 (D) 1

Open solution

Q104. JEE Main 2022 (Online) 26th July Morning Shift

In Young's double slit experiment, the fringe width is $12\ \mathrm{mm}$. If the entire arrangement is placed in water of refractive index $\frac{4}{3}$, then the fringe width becomes (in mm):

(A) 16 (B) 9 (C) 48 (D) 12

Open solution

Q105. JEE Main 2022 (Online) 30th June Morning Shift

Find the ratio of maximum intensity to the minimum intensity in the interference pattern if the widths of the two slits in Young's experiment are in the ratio of 9 : 16. (Assuming intensity of light is directly proportional to the width of slits)

(A) 3 : 4 (B) 4 : 3 (C) 7 : 1 (D) 49 : 1

Open solution

Q106. JEE Main 2022 (Online) 29th June Morning Shift

Using Young's double slit experiment, a monochromatic light of wavelength 5000 $\overset{o}{A}$ produces fringes of fringe width 0.5 mm. If another monochromatic light of wavelength 6000 $\overset{o}{A}$ is used and the separation between the slits is doubled, then the new fringe width will be :

(A) 0.5 mm (B) 1.0 mm (C) 0.6 mm (D) 0.3 mm

Open solution

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Q107. JEE Main 2022 (Online) 28th June Evening Shift

In Young's double slit experiment performed using a monochromatic light of wavelength λ, when a glass plate (μ = 1.5) of thickness xλ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be :

Open solution

(A) 3 (B) 2 (C) 1.5 (D) 0.5

Q108. JEE Main 2022 (Online) 28th June Morning Shift

The aperture of the objective is 24.4 cm. The resolving power of this telescope, if a light of wavelength 2440 $\overset{o}{A}$ is used to see the object will be :

Open solution

(A) 8.1×10^6 (B) 10.0×10^7 (C) 8.2×10^5 (D) 1.0×10^{-8}

Q109. JEE Main 2022 (Online) 26th June Evening Shift

For a specific wavelength 670 nm of light coming from a galaxy moving with velocity v, the observed wavelength is 670.7 nm. The value of v is :

Open solution

(A) 3×10^8 ms^{-1} (B) 3×10^{10} ms^{-1}

(C) 3.13×10^5 ms^{-1} (D) 4.48×10^5 ms^{-1}

Q110. JEE Main 2022 (Online) 25th June Evening Shift

The interference pattern is obtained with two coherent light sources of intensity ratio 4 : 1. And the ratio $\frac{I_{max}+I_{min}}{I_{max}-I_{min}}$ is $\frac{5}{x}$. Then, the value of x will be equal to :

Open solution

(A) 3 (B) 4 (C) 2 (D) 1

Q111. JEE Main 2022 (Online) 25th June Evening Shift

A light whose electric field vectors are completely removed by using a good polaroid, allowed to incident on the surface of the prism at Brewster's angle. Choose the most suitable option for the phenomenon related to the prism.

Open solution

(A) Reflected and refracted rays will be perpendicular to each other.

(B) Wave will propagate along the surface of prism.

(C) No refraction, and there will be total reflection of light.

(D) No reflection, and there will be total transmission of light.

Q112. JEE Main 2022 (Online) 25th June Morning Shift

The two light beams having intensities I and 9I interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be :

Open solution

(A) 2 I (B) 6 I (C) 5 I (D) 7 I

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Q113. JEE Main 2022 (Online) 24th June Evening Shift

Two light beams of intensities in the ratio of 9 : 4 are allowed to interfere. The ratio of the intensity of maxima and minima will be :

(A) 2 : 3 (B) 16 : 81 (C) 25 : 169 (D) 25 : 1

Open solution

Q114. JEE Main 2021 (Online) 27th August Evening Shift

The light waves from two coherent sources have same intensity $I_1 = I_2 = I_0$. In interference pattern the intensity of light at minima is zero. What will be the intensity of light at maxima?

(A) I_0 (B) $2\ I_0$ (C) $5\ I_0$ (D) $4\ I_0$

Open solution

Q115. JEE Main 2021 (Online) 27th July Morning Shift

In Young's double slit experiment, if the source of light changes from orange to blue then :

(A) the central bright fringe will become a dark fringe.

(B) the distance between consecutive fringes will decrease.

(C) the distance between consecutive fringes will increases.

(D) the intensity of the minima will increase.

Open solution

Q116. JEE Main 2021 (Online) 25th July Morning Shift

In the Young's double slit experiment, the distance between the slits varies in time as $d(t) = d_0 + a_0 \sin\omega t$; where d_0, ω and a_0 are constants. The difference between the largest fringe width and the smallest fringe width obtained over time is given as :

(A) $\frac{2\lambda D(d_0)}{(d_0^2 - a_0^2)}$ (B) $\frac{2\lambda D a_0}{(d_0^2 - a_0^2)}$

(C) $\frac{\lambda D}{d_0^2} a_0$ (D) $\frac{\lambda D}{d_0 + a_0}$

Open solution

Q117. JEE Main 2021 (Online) 20th July Evening Shift

With what speed should a galaxy move outward with respect to earth so that the sodium-D line at wavelength 5890 $\overset{o}{A}$ is observed at 5896 $\overset{o}{A}$?

(A) 306 km/sec (B) 322 km/sec (C) 296 km/sec (D) 336 km/sec

Open solution

Q118. JEE Main 2021 (Online) 18th March Morning Shift

In Young's double slit arrangement, slits are separated by a gap of 0.5 mm, and the screen is placed at a distance of 0.5 m from them. The distance between the first and the third bright fringe formed when the slits are illuminated by a monochromatic light of 5890 $\overset{o}{A}$ is :-

(A) 1178×10^{-9} m (B) 1178×10^{-6} m (C) 1178×10^{-12} m (D) 5890×10^{-7} m

Open solution

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Q119. JEE Main 2021 (Online) 26th February Morning Shift

In a Young's double slit experiment two slits are separated by 2 mm and the screen is placed one meter away. When a light of wavelength 500 nm is used, the fringe separation will be :

(A) 0.50 mm (B) 0.25 mm (C) 1 mm (D) 0.75 mm

Open solution

Q120. JEE Main 2021 (Online) 25th February Evening Shift

Consider the diffraction pattern obtained from the sunlight incident on a pinhole of diameter 0.1 μm. If the diameter of the pinhole is slightly increased, it will affect the diffraction pattern such that:

(A) its size increases, but intensity decreases

(B) its size increases, and intensity increases

(C) its size decreases, but intensity increases

(D) its sizes decreases, and intensity decreases

Open solution

Q121. JEE Main 2021 (Online) 25th February Morning Shift

Two coherent light sources having intensity in the ratio 2x produce an interference pattern. The ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ will be :

(A) $\frac{2\sqrt{2x}}{2x+1}$ (B) $\frac{2\sqrt{2x}}{x+1}$ (C) $\frac{\sqrt{2x}}{x+1}$ (D) $\frac{\sqrt{2x}}{2x+1}$

Open solution

Q122. JEE Main 2021 (Online) 24th February Evening Shift

If the source of light used in a Young's double slit experiment is changed from red to violet :

(A) the fringes will become brighter.

(B) the intensity of minima will increase.

(C) consecutive fringe lines will come closer.

(D) the central bright fringe will become a dark fringe.

Open solution

Q123. JEE Main 2021 (Online) 24th February Morning Shift

In a Young's double slit experiment, the width of the one of the slit is three times the other slit. The amplitude of the light coming from a slit is proportional to the slit-width. Find the ratio of the maximum to the minimum intensity in the interference pattern.

(A) 4 : 1 (B) 2 : 1 (C) 1 : 4 (D) 3 : 1

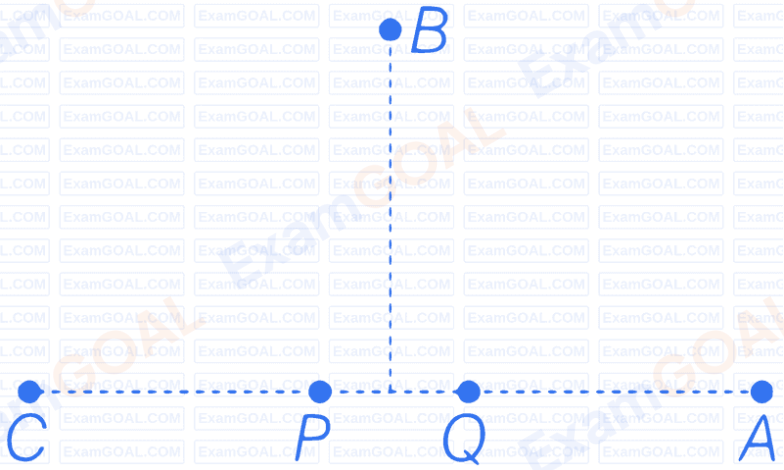
Open solution

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Q124. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

In the figure below, P and Q are two equally intense coherent sources emitting radiation of wavelength 20 m. The separation between P and Q is 5 m and the phase of P is ahead of that of Q by 90°. A, B and C are three distinct points of observation, each equidistant from the midpoint of PQ. The intensities of radiation at A, B, C will be in the ratio :



(A) 4 : 1 : 0 (B) 2 : 1 : 0 (C) 0 : 1 : 2 (D) 0 : 1 : 4

Q125. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

A beam of plane polarised light of large cross-sectional area and uniform intensity of 3.3 Wm^{-2} falls normally on a polariser (cross sectional area 3×10^{-4} m^2) which rotates about its axis with an angular speed of 31.4 rad/s. The energy of light passing through the polariser per revolution, is close to :

(A) 1.0×10^{-5} J (B) 1.0×10^{-4} J (C) 1.5×10^{-4} J (D) 5.0×10^{-4} J

Q126. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

Two light waves having the same wavelength λ in vacuum are in phase initially. Then the first wave travels a path L_1 through a medium of refractive index n_1 while the second wave travels a path of length L_2 through a medium of refractive index n_2 . After this the phase difference between the two waves is :

(A) $\frac{2\pi}{\lambda}(n_1L_1 - n_2L_2)$

(B) $\frac{2\pi}{\lambda}(n_2L_1 - n_1L_2)$

(C) $\frac{2\pi}{\lambda}\left(\frac{L_1}{n_1} - \frac{L_2}{n_2}\right)$

(D) $\frac{2\pi}{\lambda}\left(\frac{L_2}{n_1} - \frac{L_1}{n_2}\right)$

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Q127. JEE Main 2020 (Online) 3rd September Morning Slot

In a Young's double slit experiment, light of 500 nm is used to produce an interference pattern. When the distance between the slits is 0.05 mm, the angular width (in degree) of the fringes formed on the distance screen is close to

(A) 0.17^o (B) 1.7^o (C) 0.57^o (D) 0.07^o

Open solution

Q128. JEE Main 2020 (Online) 2nd September Evening Slot

In a Young's double slit experiment, 16 fringes are observed in a certain segment of the screen when light of a wavelength 700 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of the screen would be

(A) 28 (B) 24 (C) 30 (D) 18

Open solution

Q129. JEE Main 2020 (Online) 2nd September Morning Slot

Interference fringes are observed on a screen by illuminating two thin slits 1 mm apart with a light source (λ = 632.8 nm). The distance between the screen and the slits is 100 cm. If a bright fringe is observed on a screen at a distance of 1.27 mm from the central bright fringe, then the path difference between the waves, which are reaching this point from the slits is close is

(A) 1.27 μm (B) 2.05 μm (C) 2.87 nm (D) 2 nm

Open solution

Q130. JEE Main 2020 (Online) 9th January Morning Slot

The aperture diameter of a telescope is 5m. The separation between the moon and the earth is 4×10^5 km. With light of wavelength of 5500 $\overset{o}{A}$, the minimum separation between objects on the surface of moon, so that they are just resolved, is close to :

(A) 20 m (B) 200 m (C) 600 m (D) 60 m

Open solution

Q131. JEE Main 2020 (Online) 8th January Evening Slot

In a double slit experiment, at a certain point on the screen the path difference between the two interfering waves is $\frac{1}{8}$th of a wavelength. The ratio of the intensity of light at that point to that at the centre of a bright fringe is :

(A) 0.853 (B) 0.568 (C) 0.672 (D) 0.760

Open solution

Q132. JEE Main 2020 (Online) 7th January Evening Slot

In a Young's double slit experiment, the separation between the slits is 0.15 mm. in the experiment, a source of light of wavelengh 589 nm is used and the interference pattern is observed on a screen kept 1.5 m away. The separation between the successive bright fringes on the screen is :

(A) 4.9 mm (B) 5.9 mm (C) 6.9 mm (D) 3.9 mm

Open solution

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Q133. JEE Main 2020 (Online) 7th January Morning Slot

A polarizer - analyser set is adjusted such that the intensity of light coming out of the analyser is just 10% of the original intensity. Assuming that the polarizer - analyser set does not absorb any light, the angle by which the analyser need to be rotated further to reduce the output intensity to be zero, is :

Open solution

(A) 71.6^o (B) 90° (C) 18.4^o (D) 45°

Q134. JEE Main 2020 (Online) 7th January Morning Slot

Visible light of wavelength 6000×10^{-8} cm falls normally on a single slit and produces a diffraction pattern. It is found that the second diffraction minimum is at 60° from the central maximum. If the first minimum is produced at θ_1, then θ_1, is close to :

Open solution

(A) 45° (B) 30° (C) 25° (D) 20°

Q135. JEE Main 2019 (Online) 12th April Evening Slot

A system of three polarizers P_1, P_2, P_3 is set up such that the pass axis of P_3 is crossed with respect to that of P_1. The pass axis of P_2 is inclined at 60° to the pass axis of P_3. When a beam of unpolarized light of intensity I_0 is incident on P_1, the intensity of light transmitted by the three polarizers is I. The ratio $(\frac{I_0}{I})$ equals (nearly) :

Open solution

(A) 10.67 (B) 5.33 (C) 16.00 (D) 1.80

Q136. JEE Main 2019 (Online) 12th April Morning Slot

In a double slit experiment, when a thin film of thickness t having refractive index μ. is introduced in front of one of the slits, the maximum at the centre of the fringe pattern shifts by one fringe width. The value of t is (λ is the wavelength of the light used) :

Open solution

(A) $\frac{\lambda}{2(\mu-1)}$ (B) $\frac{\lambda}{(2\mu-1)}$

(C) $\frac{2\lambda}{(\mu-1)}$ (D) $\frac{\lambda}{(\mu-1)}$

Q137. JEE Main 2019 (Online) 12th April Morning Slot

The value of numerical aperature of the objective lens of a microscope is 1.25. If light of wavelength 5000 $\overset{o}{A}$ is used, the minimum separation between two points, to be seen as distinct, will be :

Open solution

(A) 0.12 μm (B) 0.38 μm (C) 0.24 μm (D) 0.48 μm

Q138. JEE Main 2019 (Online) 10th April Evening Slot

In a Young's double slit experiment, the ratio of the slit's width is 4 : 1. The ratio of the intensity of maxima to minima, close to the central fringe on the screen, will be :

Open solution

(A) 25 : 9 (B) 4 : 1

(C) $\left(\sqrt{3}+1\right)^4 : 16$ (D) 9 : 1

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Q139. JEE Main 2019 (Online) 9th April Evening Slot

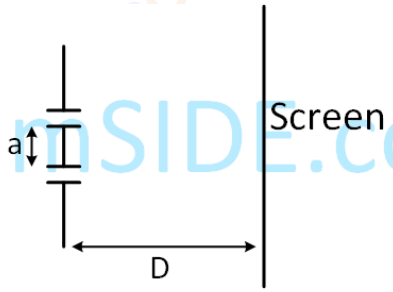
Diameter of the objective lens of a telescope is 250 cm. For light of wavelength 600nm. coming from a distant object, the limit of resolution of the telescope is close to :-

(A) 3.0×10^{-7} rad (B) 4.5×10^{-7} rad (C) 1.5×10^{-7} rad (D) 2.0×10^{-7} rad

Open solution

Q140. JEE Main 2019 (Online) 9th April Morning Slot

The figure shows a Young's double slit experimental setup. It is observed that when a thin transparent sheet of thickness t and refractive index μ is put in front of one of the slits, the central maximum gets shifted by a distance equal to n fringe widths. If the wavelength of light used is λ, t will be :



(A) $\frac{D\lambda}{a(\mu-1)}$ (B) $\frac{2nD\lambda}{a(\mu-1)}$

(C) $\frac{2D\lambda}{a(\mu-1)}$ (D) $\frac{n\lambda}{(\mu-1)}$

Open solution

Q141. JEE Main 2019 (Online) 8th April Evening Slot

Calculate the limit of resolution of a telescope objective having a diameter of 200 cm, if it has to detect light of wavelength 500 nm coming from a star :-

(A) 610×10^{-9} radian (B) 457.5×10^{-9} radian

(C) 305×10^{-9} radian (D) 152.5×10^{-9} radian

Open solution

Q142. JEE Main 2019 (Online) 8th April Morning Slot

In an interference experiment the ratio of amplitudes of coherent waves is $\frac{a_1}{a_2} = \frac{1}{3}$. The ratio of maximum and minimum intensities of fringes will be :

(A) 2 (B) 4 (C) 18 (D) 9

Open solution

Q143. JEE Main 2019 (Online) 11th January Evening Slot

In a double-slit experiment, green light ($5303\mathring{A}$) falls on a double slit having a separation of 19.44 μm and a width of 4.05 μm. The number of bright fringes between the first and the second diffraction minima is :

(A) 04 (B) 05 (C) 10 (D) 09

Open solution

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Q144. JEE Main 2019 (Online) 11th January Morning Slot

In a Young's double slit experiment, the path difference, at a certain point on the screen, between two interfering waves is $\frac{1}{8}$ th of wavelength. The ratio of the intensity at this point to that at the centre of a bright fringe is close to :

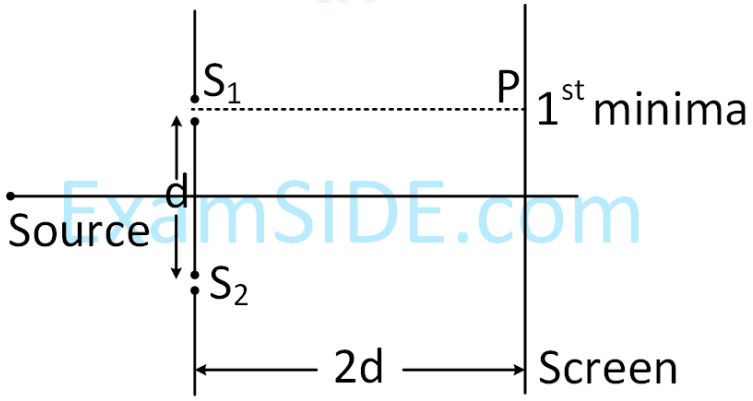
Open solution

(A) 0.94 (B) 0.85 (C) 0.74 (D) 0.80

Q145. JEE Main 2019 (Online) 10th January Evening Slot

Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first minima occurs directly in front of the slit (S_1) ?

Open solution



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(A) $\frac{\lambda}{2\left(5-\sqrt{2}\right)}$ (B) $\frac{\lambda}{2\left(\sqrt{5}-2\right)}$

(C) $\frac{\lambda}{\left(5-\sqrt{2}\right)}$ (D) $\frac{\lambda}{\left(\sqrt{5}-2\right)}$

Q146. JEE Main 2019 (Online) 10th January Morning Slot

In a Young's double slit experiment with slit separation 0.1 mm, one observes a bright fringe at angle $\frac{1}{40}$ by using light of wavelength λ_1. When the light of wavelength λ_2 is used a bright fringe is seen at the same angle in the same set up. Given that λ_1 and λ_2 are in visible range (380 nm to 740 nm), their values are -

Open solution

(A) 400 nm, 500 nm (B) 625 nm, 500 nm (C) 380 nm, 500 nm (D) 380 nm, 525 nm

Q147. JEE Main 2019 (Online) 9th January Evening Slot

In a Young's double slit experiment, the slits are placed 0.320 mm apart. Light of wavelength λ = 500 nm is incident on the slits. The total number of bright fringes that are observed in the angular range $-30^{o} \leq \theta \leq 30^{o}$ is :

Open solution

(A) 640 (B) 320 (C) 321 (D) 641

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Q148. JEE Main 2019 (Online) 9th January Morning Slot

Two coherent sources produce waves of different intensities which interfere. After interference, the ratio of the maximum intensity to the minimum intensity is 16. The intensity of the waves are in the ratio :

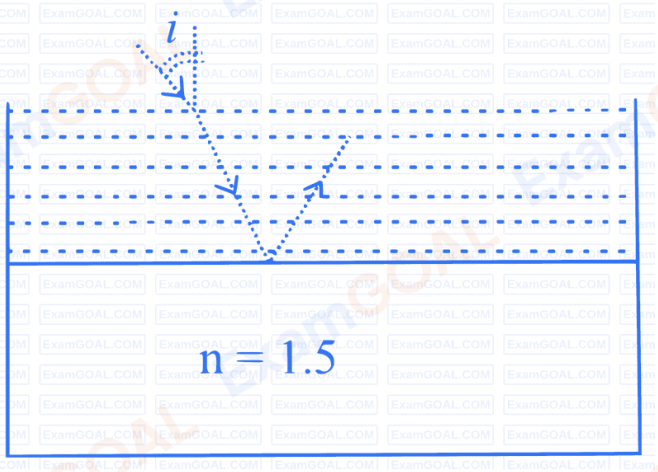
Open solution

(A) 16 : 9 (B) 25 : 9 (C) 4 : 1 (D) 5 : 3

Q149. JEE Main 2019 (Online) 9th January Morning Slot

Consider a tank made of glass(refractive index 1.5) with a thick bottom. It is filled with a liquid of refractive index μ. A student finds that, irrespective of what the incident angle i (see figure) is for a beam of light entering the liquid, the light reflected from the liquid glass interface is never completely polarized. For this to happen, the minimum value of μ is :

Open solution



(A) $\sqrt{\frac{5}{3}}$ (B) $\frac{3}{\sqrt{5}}$ (C) $\frac{5}{\sqrt{3}}$ (D) $\frac{4}{3}$

Q150. JEE Main 2018 (Online) 16th April Morning Slot

Unpolarized light of intensity I is incident on a system of two polarizers, A followed by B. The intensity of emergent light is I/2. If a third polarizer C is placed between A and B, the intensity of emergent light is reduced to I/3. The angle between the polarizers A and C is θ. Then :

Open solution

(A) $\cos\theta = \left(\frac{2}{3}\right)^{1/2}$

(B) $\cos\theta = \left(\frac{2}{3}\right)^{1/4}$

(C) $\cos\theta = \left(\frac{1}{3}\right)^{1/2}$

(D) $\cos\theta = \left(\frac{1}{3}\right)^{1/4}$

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Q151. JEE Main 2018 (Offline)

The angular width of the central maximum in a single slit diffraction pattern is 60°. The width of the slit is 1 μm. The slit is illuminated by monochromatic plane waves. If another slit of same width is made near it, Young's fringes can be observed on a screen placed at a distance 50 cm from the slits. If the observed fringe width is 1 cm, what is slit separation distance? (i.e. distance between the centres of each slit.)

Open solution

(A) 100 μm (B) 25 μm (C) 50 μm (D) 75 μm

Q152. JEE Main 2018 (Offline)

Unpolarized light of intensity I passes through an ideal polarizer A. Another identical polarizer B is placed behind A. The intensity of light beyond B is found to be I/2. Now another identical polarizer C is placed between A and B. The intensity beyond B is now found to be I/8. The angle between polarizer A and C is :

Open solution

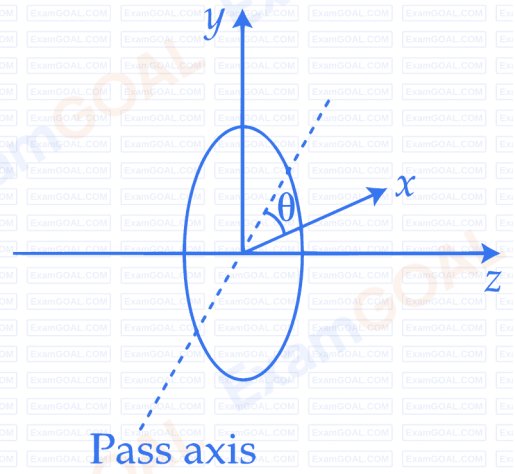
(A) 60° (B) 30° (C) 45° (D) 0°

Q153. JEE Main 2018 (Online) 15th April Evening Slot

A plane polarized light is incident on a polariser with its pass axis making angle θ with x-axis, as shown in the figure. At four different values of θ, $\theta = 8^o, 38^o, 188^o$ and 218°, the observed intensities are same.

What is the angle between the direction of polarization and x-axis ?

Open solution



(A) 98° (B) 128° (C) 203° (D) 45°

Q154. JEE Main 2018 (Online) 15th April Morning Slot

Light of wavelength $550\,nm$ falls normally on a slit of width $22.0 \times 10^{-5}\,cm$. The angular position of the second minima from the central maximum will be (in radians) :

Open solution

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{8}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

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Q155. JEE Main 2017 (Online) 9th April Morning Slot

A single slit of width 0.1 mm is illuminated by a parallel beam of light of wavelength 6000 $\mathring{A}$ and diffraction bands are observed on a screen 0.5 m from the slit. The distance of the third dark band from the central bright band is :

Open solution

(A) 3 mm (B) 9 mm (C) 4.5 mm (D) 1.5 mm

Q156. JEE Main 2017 (Online) 8th April Morning Slot

A single slit of width b is illuminated by a coherent monochromatic light of wavelength λ. If the second and fourthminima in the diffraction pattern at a distance 1 m from the slit are at 3 cm and 6 cm respectively from the central maximum, what is the width of the central maximum ? (i.e. distance between first minimum on either side of the central maximum)

Open solution

(A) 1.5 cm (B) 3.0 cm (C) 4.5 cm (D) 6.0 cm

Q157. JEE Main 2017 (Offline)

In a Young's double slit experiment, slits are separated by 0.5 mm, and the screen is placed 150 cm away. A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes on the screen. The least distance from the common central maximum to the point where the bright fringes due to both the wavelengths coincide is

Open solution

(A) 15.6 mm (B) 1.56 mm (C) 7.8 mm (D) 9.75 mm

Q158. JEE Main 2016 (Online) 10th April Morning Slot

Two stars are 10 light years away from the earth. They are seen through a telescope of objective diameter 30 cm. The wavelength of light is 600 nm. To see the stars just resolved by the telescope, the minimum distance between them should be (1 light year = 9.46×10^{15} m) of the order of :

Open solution

(A) 10^6 km (B) 10^8 km (C) 10^{11} km (D) 10^{10} km

Q159. JEE Main 2016 (Online) 9th April Morning Slot

In Young's double slit experiment, the distance between slits and the screen is 1.0 m and monochromatic light of 600 nm is being used. A person standing near the slits is looking at the fringe pattern. When the separation between the slits is varied, the interference pattern disappears for a particular distance d_0 between the slits. If the angular resolution of the eye is $(\frac{1}{60})^o$, the value of d_0 is close to :

Open solution

(A) 1 mm (B) 2 mm (C) 4 mm (D) 3 mm

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Q160. JEE Main 2016 (Offline)

The box of a pin hole camera, of length L, has a hole of radius a. It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say $b_{\min}$) when :

(A) $a = \sqrt{\lambda L}$ and $b_{\min} = \sqrt{4\lambda L}$

(B) $a = \frac{\lambda^2}{L}$ and $b_{\min} = \sqrt{4\lambda L}$

(C) $a = \frac{\lambda^2}{L}$ and $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$

(D) $a = \sqrt{\lambda L}$ and $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$

Open solution

Q161. JEE Main 2015 (Online) 11th April Morning Slot

In a Young's double slit experiment with light of wavelength λ the separation of slits is d and distance of screen is D such that D > > d > > λ. If the Fringe width is β, the distance from point of maximum intensity to the point where intensity falls to half of maximum intensity on either side is :

(A) $\frac{\beta}{2}$ (B) $\frac{\beta}{4}$ (C) $\frac{\beta}{3}$ (D) $\frac{\beta}{6}$

Open solution

Q162. JEE Main 2015 (Online) 11th April Morning Slot

Unpolarized light of intensity I_0 is incident on surface of a block of glass at Brewster's angle . In that case, which one of the following statements is true ?

(A) transmitted light is partially polarized with intensity $I_0/2$.

(B) transmitted light is completely polarized with intensity $I_0/2$.

(C) reflected light is completely polarized with intensity less than $I_0/2$.

(D) reflected light is partially polarized with intensity $I_0/2$.

Open solution

Q163. JEE Main 2015 (Offline)

Assuming human pupil to have a radius of $0.25\ cm$ and a comfortable viewing distance of $25\ cm$, the minimum separation between two objects that human eye can resolve at $500\ nm$ wavelength is :

(A) $100\ \mu m$ (B) $300\ \mu m$ (C) $1\ \mu m$ (D) $30\ \mu m$

Open solution

Q164. JEE Main 2015 (Offline)

On a hot summer night, the refractive index of air is smallest near the ground and increases with height from the ground. When a light beam is directed horizontally, the Huygens' principle leads us to conclude that as it travels, the light beam :

(A) bends down wards

(B) bends upwards

(C) becomes narrower

(D) goes horizontally without any deflection

Open solution

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Q165. JEE Main 2014 (Online) 19th April Morning Slot

A ray of light is incident from a denser to a rarer medium. The critical angle for total internal reflection is θ_{iC} and the Brewster's angle of incidence is θ_{iB}, such that $\sin\theta_{iC}/\sin\theta_{iB} = \eta = 1.28$. The relative refractive index of the two media is :

Open solution

(A) 0.2 (B) 0.4 (C) 0.8 (D) 0.9

Q166. JEE Main 2014 (Online) 19th April Morning Slot

The diameter of the objective lens of microscope makes an angle β at the focus of the microscope. Further, the medium between the object and the lens is an oil of refractive index n. Then the resolving power of the microscope.

Open solution

(A) Increases with decreasing value of n

(B) Increases with decreasing value of β

(C) Increases with increasing value of n sin 2β

(D) Increases with increasing value of $\frac{1}{n\sin 2\beta}$

Q167. JEE Main 2014 (Online) 19th April Morning Slot

In a Young's double slit experiment, the distance between the two identical slits is 6.1 times larger than the slit width. Then the number of intensity maxima observed within the central maximum of the single slit diffraction pattern is :

Open solution

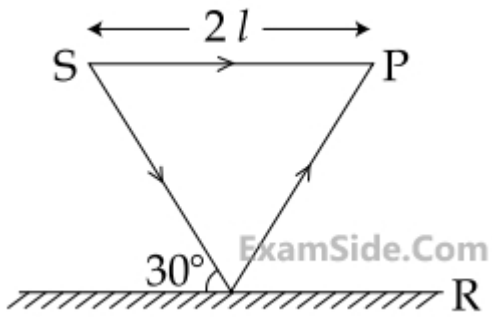
(A) 3 (B) 6 (C) 12 (D) 24

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Q168. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

Interference pattern is observed at 'P' due to superimposition of two rays coming out from a source 'S' as shown in the figure. The value of $'l'$ for which maxima is obtained at 'P' is :
(R is perfect reflecting surface) :



(A) $l = \frac{2n\lambda}{\sqrt{3}-1}$

(B) $l = \frac{(2n-1)\lambda}{2\left(\sqrt{3}-1\right)}$

(C) $l = \frac{(2n-1)\lambda\sqrt{3}}{4\left(2-\sqrt{3}\right)}$

(D) $l = \frac{(2n-1)\lambda}{\sqrt{3}-1}$

Q169. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

In an experiment of single slit diffraction pattern, first minimum for red light coincides with first maximum of some other wavelength. If wavelength of red light is 6600 $\mathring{A}$, then wavelength of first maximum will be :

(A) 3300 $\mathring{A}$

(B) 4400 $\mathring{A}$

(C) 5500 $\mathring{A}$

(D) 6600 $\mathring{A}$

Q170. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

Two monochromatic light beams of intensity 16 and 9 units are interfering. The ratio of intensities of bright and dark parts of the resultant pattern is :

(A) $\frac{16}{9}$

(B) $\frac{4}{3}$

(C) $\frac{7}{1}$

(D) $\frac{49}{1}$

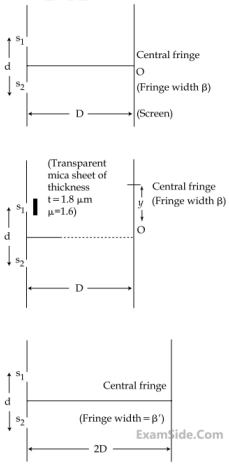
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Q171. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

Using monochromatic light of wavelength λ, an experimentalist sets up the Young's double slit experiment in three ways as shown.

If she observes that y = β', the wavelength of light used is :



(A) 520 nm (B) 540 nm (C) 560 nm (D) 580 nm

Q172. JEE Main 2014 (Offline)

Open solution

Two beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through 30° makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and I_B respectively, then $\frac{\mathrm{I}_A}{\mathrm{I}_B}$ equals:

(A) 3 (B) $\frac{3}{2}$ (C) 1 (D) $\frac{1}{3}$

Q173. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

A thin glass plate of thickness $\frac{2500}{3}\lambda$ (λ is wavelength of light used) and refractive index $\mu = 1.5$ is inserted between one of the slits and the screen in Young's double slit experiment. At a point on the screen equidistant from the slits, the ratio of the intensities before and after the introduction of the glass plate is :

(A) 2 : 1 (B) 1 : 4 (C) 4 : 1 (D) 4 : 3

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Q174. JEE Main 2013 (Online) 25th April Morning Slot

The source that illuminates the double-slit in 'double-slit interference experiment' emits two distinct monochromatic waves of wavelength 500 nm and 600 nm, each of them producing its own pattern on the screen. At the central point of the pattern when path difference is zero, maxima of both the patterns coincide and the resulting interference pattern is most distinct at the region of zero path difference. But as one moves out of this central region, the two fringe systems are gradually out of step such that maximum due to one wavelength coincides with the minimum due to the other and the combined fringe system becomes completely indistinct. This may happen when path difference in nm is:

Open solution

(A) 2000 (B) 3000 (C) 1000 (D) 1500

Q175. JEE Main 2013 (Online) 23th April Morning Slot

n identical waves each of intensity I_0 interfere with each other. The ratio of maximum intensities if the interference is (i) coherent and (ii) incoherent is :

Open solution

(A) n^2 (B) $\frac{1}{n}$ (C) $\frac{1}{n^2}$ (D) n

Q176. JEE Main 2013 (Online) 22th April Morning Slot

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement-1: In Young's double slit experiment, the number of fringes observed in the field of view is small with longer wavelength of light and is large with shorter wavelength of light.
Statement-2: In the double slit experiment the fringe width depends directly on the wavelength of light.

Open solution

(A) Statement-1 is true, Statement-2 is true and the Statement-2 is correct explanation of the Statement-1.

(B) Statement-1 is false and the Statement-2 is true.

(C) Statement-1 is true Statement-2 is true and the Statement-2 is not correct explanation of the Statement-1.

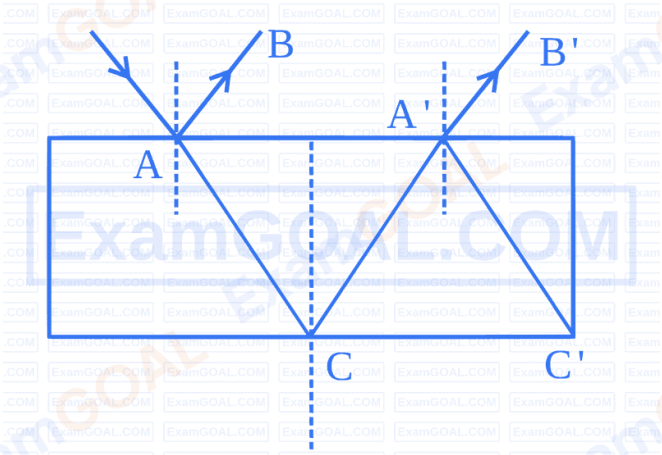
(D) Statement-1 is true and the Statement-2 is false.

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Q177. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

A ray of light of intensity I is incident on a parallel glass slab at point A as shown in the diagram. It undergoes partial reflection and refraction. At each reflection, 25% of incident energy is reflected. The rays AB and A'B' undergo interference. The ratio of I_{max} and I_{min} is :

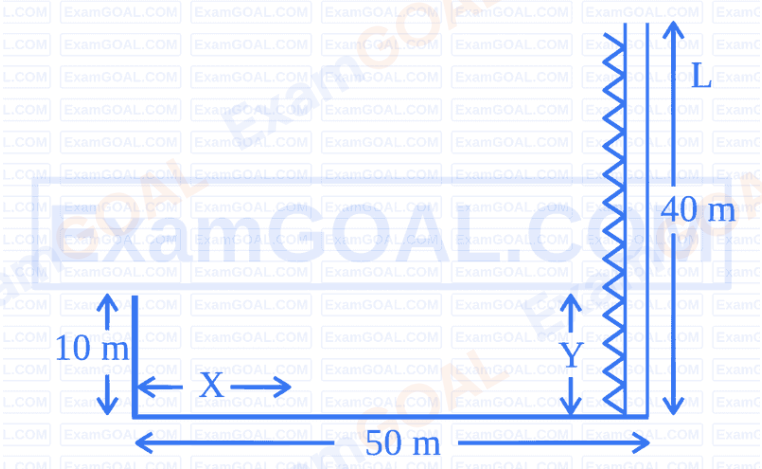


(A) 49 : 1 (B) 7 : 1 (C) 4 : 1 (D) 8 : 1

Q178. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

A person lives in a high-rise building on the bank of a river 50 m wide. Across the river is a well lit tower of height 40 m. When the person, who is at a height of 10 m, looks through a polarizer at an appropriate angle at light of the tower reflecting from the river surface, he notes that intensity of light coming from distance X from his building is the least and this corresponds to the light coming from light bulbs at height 'Y' on the tower. The values of X and Y are respectively close to (refractive index of water $\sim \frac{4}{3}$)



(A) 25 m, 10 m (B) 13 m, 27 m (C) 22 m, 13 m (D) 17 m, 20 m

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Q179. JEE Main 2013 (Offline)

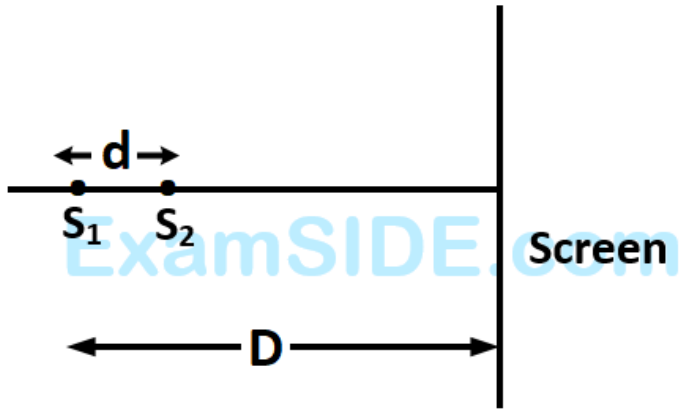
A beam of unpolarised light of intensity I_0 is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of 45° relative to that of A. The intensity of the emergent light is

(A) I_0 (B) $\frac{I_0}{2}$ (C) $\frac{I_0}{4}$ (D) $\frac{I_0}{8}$

Open solution

Q180. JEE Main 2013 (Offline)

Two coherent point sources S_1 and S_2 are separated by a small distance $'d'$ as shown. The fringes obtained on the screen will be



(A) points
(B) straight lines
(C) semi-circles
(D) concentric circles

Open solution

Q181. AIEEE 2012 (Online) 26th May

A telescope of aperture 3×10^{-2} m diameter is focused on a window at 80 m distance fitted with a wire mesh of spacing 2×10^{-3} m. Given: $\lambda = 5.5 \times 10^{-7}$ m, which of the following is true for observing the mesh through the telescope?

(A) Yes, it is possible with the same aperture size.
(B) Possible also with an aperture half the present diameter.
(C) No, it is not possible.
(D) Given data is not sufficient.

Open solution

Q182. AIEEE 2012 (Online) 26th May

In Young's double slit interference experiment, the slit widths are in the ratio $1 : 25$. Then the ratio of intensity at the maxima and minima in the interference pattern is

(A) $3 : 2$ (B) $1 : 25$ (C) $9 : 4$ (D) $1 : 5$

Open solution

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Q183. AIEEE 2012 (Online) 19th May

Two coherent plane light waves of equal amplitude makes a small angle α ($\ll 1$) with each other. They fall almost normally on a screen. If λ is the wavelength of light waves, the fringe width Δx of interference patterns of the two sets of waves on the screen is

(A) $\frac{2\lambda}{\alpha}$ (B) $\frac{\lambda}{\alpha}$ (C) $\frac{\lambda}{(2\alpha)}$ (D) $\frac{\lambda}{\sqrt{\alpha}}$

Open solution

Q184. AIEEE 2012 (Online) 12th May

The first diffraction minimum due to the single slit diffraction is seen at $\theta = 30^\circ$ for a light of wavelength 5000 Å falling perpendicularly on the slit. The width of the slit is

(A) 2.5×10^{-5} cm (B) 1.25×10^{-5} cm (C) 10×10^{-5} cm (D) 5×10^{-5} cm

Open solution

Q185. AIEEE 2012 (Online) 12th May

The maximum number of possible interference maxima for slit separation equal to 1.8λ, where λ is the wavelength of light used, in a Young's double slit experiment is

(A) zero (B) 3 (C) infinite (D) 5

Open solution

Q186. AIEEE 2012 (Online) 7th May

In a Young's double slit experiment with light of wavelength λ, fringe pattern on the screen has fringe width β. When two thin transparent glass (refractive index μ) plates of thickness t_1 and t_2 ($t_1 > t_2$) are placed in the path of the two beams respectively, the fringe pattern will shift by a distance

(A) $\frac{\beta(\mu-1)}{\lambda}\left(\frac{t_1}{t_2}\right)$ (B) $\frac{\mu\beta}{\lambda}\frac{t_1}{t_2}$

(C) $\frac{\beta(\mu-1)}{\lambda}(t_1 - t_2)$ (D) $(\mu-1)\frac{\lambda}{\beta}(t_1 + t_2)$

Open solution

Q187. AIEEE 2012 (Online) 7th May

Two polaroids have their polarizing directions parallel so that the intensity of a transmitted light is maximum. The angle through which either polaroid must be turned if the intensity is to drop by one-half is

(A) 135° (B) 90° (C) 120° (D) 180°

Open solution

Q188. AIEEE 2012

In Young's double slit experiment , one of the slit is wider than other, so that amplitude of the light from one slit is double of that from other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at phase difference ϕ is given by :

(A) $\frac{I_m}{9}(4 + 5\cos\phi)$ (B) $\frac{I_m}{3}\left(1 + 2\cos^2\frac{\phi}{2}\right)$

(C) $\frac{I_m}{3}\left(1 + 4\cos^2\frac{\phi}{2}\right)$ (D) $\frac{I_m}{9}\left(1 + 8\cos^2\frac{\phi}{2}\right)$

Open solution

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Q189. AIEEE 2011

Open solution

This question has a paragraph followed by two statements, Statement -1 and Statement -2. Of the given four alternatives after the statements, choose the one that describes the statements.
A thin air film is formed by putting the convex surface of a plane-convex lens over a plane glass plane. With monochromatic light, this film gives an interference pattern due to light, reflected from the top (convex) surface and the bottom (glass plate) surface of the film.

Statement - 1 **:** When light reflects from the air-glass plate interface, the reflected wave suffers a phase change of π.

Statement - 2 **:** The center of the interference pattern is dark.

(A) Statement - 1 is true, Statement - 2 is true, Statement - 2 is the correct explanation of Statement - 1

(B) Statement - 1 is true, Statement - 2 is true, Statement - 2 is not the correct explanation of Statement - 1

(C) Statement - 1 is false, Statement - 2 is true

(D) Statement - 1 is true, Statement - 2 is false.

Q190. AIEEE 2010

Open solution

An initially parallel cylindrical beam travels in a medium of refractive index $\mu(I) = \mu_0 + \mu_2 I$, where μ_0 and μ_2 are positive constants and I is the intensity of the light beam. The intensity of the beam is decreasing with increasing radius.
The initial shape of the wavefront of the beam is

(A) convex

(B) concave

(C) convex near the axis and concave near the periphery

(D) planar

Q191. AIEEE 2009

Open solution

A mixture of light, consisting of wavelength $590\,nm$ and an unknown wavelength, illuminates Young's double slit and gives rise to two overlapping interference patterns on the screen. The central maximum of both lights coincide. Further, it is observed that the third bright fringe of known light coincides with the 4th bright fringe of the unknown light. From this data, the wavelength of the unknown light is :

(A) $885.0\,nm$ (B) $442.5\,nm$ (C) $776.8\,nm$ (D) $393.4\,nm$

Q192. AIEEE 2007

Open solution

In a Young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of light used) is I. If I_0 denotes the maximum intensity, $\frac{I}{I_0}$ is equal to

(A) $\frac{3}{4}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

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Q193. AIEEE 2005

Two point white dots are $1\ mm$ apart on a black paper. They are viewed by eye of pupil diameter $3\ mm$. Approximately, what is the maximum distance at which these dots can be resolved by the eye? [Take wavelength of light $= 500\ nm$]

(A) $1m$ (B) $5m$ (C) $3m$ (D) $6m$

Open solution

Q194. AIEEE 2005

A Young's double slit experiment uses a monochromatic source. The shape of the interference fringes formed on a screen is

(A) circle (B) hyperbola (C) parabola (D) straight line

Open solution

Q195. AIEEE 2005

If I_0 is the intensity of the principal maximum in the single slit diffraction pattern, then what will be its intensity when the slit width is doubled?

(A) $4I_0$ (B) $2I_0$ (C) $\frac{I_0}{2}$ (D) I_0

Open solution

Q196. AIEEE 2005

When an unpolarized light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is

(A) $\frac{1}{4}\ I_0$ (B) $\frac{1}{2}\ I_0$ (C) I_0 (D) zero

Open solution

Q197. AIEEE 2004

The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refractive index n) is :

(A) $\tan^{-1}(1/n)$ (B) $\sin^{-1}(1/n)$
(C) $\sin^{-1}(n)$ (D) $\tan^{-1}(n)$

Open solution

Q198. AIEEE 2004

The maximum number of possible interference maxima for slit-separation equal to twice the wavelength in Young's double-slit experiment, is :

(A) three (B) five (C) infinite (D) zero

Open solution

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Q199. AIEEE 2003

To demonstrate the phenomenon of interference, we require two sources which emit radiation

(A) of nearly the same frequency

(B) of the same frequency

(C) of different wavelengths

(D) of the same frequency and having a definite phase relationship

Open solution

Q200. AIEEE 2002

Wavelengths of light used in an optical instrument are $\lambda_1 = 4000\mathring{A}$ and $\lambda_2 = 5000\mathring{A}$, then ratio of their respective resolving powers (corresponding to λ_1 and λ_2) is :

(A) $16 : 25$ (B) $9 : 1$ (C) $4 : 5$ (D) $5 : 4$

Open solution

Out of syllabus 1 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q201. JEE Main 2026 (Online) 4th April Morning Shift · MCQ (Single Correct Answer)

A telescope with objective diameter R is used to observe a distant star emitting light of wavelength 500 nm , at a resolution of 5×10^{-7} radian. The value of R is

____ cm .

(A) 61 (B) 122 (C) 244 (D) 305

Open solution

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Answer Key

Numerical

1. **203**	2. **30**	3. **5625**	4. **6**	5. **36**
6. **429**	7. **4**	8. **14**	9. **15**	10. **24**
11. **6**	12. **54**	13. **11**	14. **4**	15. **200**
16. **2**	17. **6**	18. **8**	19. **275**	20. **10**
21. **9**	22. **125**	23. **13**	24. **2**	25. **0.20**
26. **30**	27. **30 or 60**	28. **462**	29. **4**	30. **48**
31. **3**	32. **10**	33. **7**	34. **24**	35. **24**
36. **630**	37. **2**	38. **600**	39. **4**	40. **450**
41. **3**	42. **1**	43. **5**	44. **30**	45. **300**
46. **2**	47. **6**	48. **600**	49. **75**	50. **9**
51. **200**	52. **750**			

MCQ (Single Correct Answer)

53. **C**	54. **C**	55. **B**	56. **B**	57. **C**
58. **C**	59. **A**	60. **C**	61. **B**	62. **D**
63. **C**	64. **C**	65. **C**	66. **B**	67. **D**
68. **A**	69. **C**	70. **B**	71. **B**	72. **D**
73. **B**	74. **B**	75. **C**	76. **D**	77. **C**
78. **D**	79. **B**	80. **C**	81. **C**	82. **C**
83. **D**	84. **A**	85. **B**	86. **C**	87. **C**
88. **A**	89. **A**	90. **C**	91. **D**	92. **B**
93. **D**	94. **D**	95. **D**	96. **B**	97. **C**
98. **A**	99. **A**	100. **D**	101. **D**	102. **C**
103. **B**	104. **B**	105. **D**	106. **D**	107. **B**
108. **C**	109. **C**	110. **B**	111. **D**	112. **B**
113. **D**	114. **D**	115. **B**	116. **B**	117. **A**
118. **B**	119. **B**	120. **C**	121. **A**	122. **C**
123. **A**	124. **B**	125. **B**	126. **A**	127. **C**
128. **A**	129. **A**	130. **D**	131. **A**	132. **B**
133. **C**	134. **C**	135. **A**	136. **D**	137. **C**
138. **D**	139. **A**	140. **D**	141. **C**	142. **B**

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143. **B**	144. **B**	145. **B**	146. **B**	147. **D**
148. **B**	149. **B**	150. **B**	151. **B**	152. **C**
153. **C**	154. **C**	155. **B**	156. **B**	157. **C**
158. **B**	159. **B**	160. **A**	161. **B**	162. **C**
163. **D**	164. **B**	165. **C**	166. **C**	167. **C**
168. **C**	169. **B**	170. **D**	171. **B**	172. **D**
173. **C**	174. **D**	175. **D**	176. **A**	177. **A**
178. **B**	179. **C**	180. **D**	181. **A**	182. **C**
183. **B**	184. **C**	185. **B**	186. **C**	187. **A**
188. **D**	189. **B**	190. **D**	191. **B**	192. **A**
193. **B**	194. **D**	195. **D**	196. **B**	197. **D**
198. **B**	199. **D**	200. **D**		

Out of syllabus

201. **B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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